07/16/2026

Test 1 Review

NAME: ______________________________

Q.1. Simplify and then rationalize the denominator: $\sqrt[5]{\dfrac{8x^3}{y^4}}\,\sqrt[5]{\dfrac{4x^4}{y^2}}$

Q.2. Rationalize the numerator:

$$\frac{\sqrt{x^5}\,\sqrt[3]{y^5}}{y}$$

07/16/2026

Q.3. Factor $12x^2 - x - 6$ and then solve $12x^2 - x - 6 = 0$.

Q.4. Factor $3x^3 - 4x^2 - 27x + 36$ and then solve $3x^3 - 4x^2 - 27x + 36 = 0$.

Q.5. Simplify the following into one rational expression:

$$\frac{t}{t+3} + \frac{4t}{t-3} - \frac{18}{t^2-9}$$

Q.6. Rationalize the denominator:

$$\frac{\sqrt{t}-4}{\sqrt{t}+4}$$

Q.7.

32 Filling a swimming pool With water from one hose, a swimming pool can be filled in 8 hours. A second, larger hose used alone can fill the pool in 5 hours. How long would it take to fill the pool if both hoses were used simultaneously?

Q.8.

15 Preparing an alloy British sterling silver is a copper-silver alloy that is 7.5% copper by weight. How many grams of pure copper and how many grams of British sterling silver should be used to prepare 200 grams of a copper-silver alloy that is 10% copper by weight?

Q.9. Simplify the quotient of complex numbers $\frac{1-7i}{6-2i}$ into a single complex number (i.e. $a+bi$ where a, b are real).

Q.10. Find all real numbers x, y that satisfy the equation: $8+(3x+y)i = 2x-4i$.

Q.11. Find all real numbers satisfying the inequality: $1 < |x| < 4$.

Q.12. Find all real numbers satisfying the equation: $6u^{-\frac{1}{2}} - 13u^{-\frac{1}{4}} + 6 = 0$.

07/16/2026

Q.13. Find all real numbers satisfying the inequality: $(x-4)^2 > x(x+12)$.

Q.14. Find all real numbers satisfying the inequality:

$$\frac{x^2-x-2}{x^2+4x+3} \geq 0$$